

General rules

- ☐ You are expected to work on your project activities in the lab during the Thursday lab session.
(Although you may work on the project on other days as well)
- ☐ Each team will comprise 3 students mandatorily.
- ☐ There are 2 ways in which a team can be allocated a project.
 1. Each team will be allocated a project from a predefined [list of projects](#). based on their preference order. You are allowed to modify the problem statements given that the overall difficulty and aim remains the same. A form will be shared to choose the order of preferences.
OR
 2. Students who would like to implement their own project ideas may do so with approval. However, they will also be required to fill in the preference form in case their proposal is rejected.
- ☐ Based on the allocated/chosen project, each team must prepare a comprehensive proposal that justifies the minimum requirements and the possible improvisations that can be made and submit it by **8th March 2025**. [Link to Project Proposal Template](#)
- ☐ Students who are implementing their own projects, therefore, **must submit two proposals**- 1) one for the independent project, and in the case that it is rejected
2) one for the allocated project.
- ☐ Project components will be handed over once these are procured. Project can be carried during the lab sessions and outside the lab as well. **Any materials needed from the lab should be clearly mentioned in the comprehensive proposal.**

Project Requirements

- ☐ An end-to-end implementation of an IoT system including hardware for sensing of physical quantity/quantities and transmitting the sensed data using a suitable set of communication protocols to a cloud platform (Thing Speak, Blynk.io etc.).
- ☐ Prototype with working hardware on breadboard is **mandatory**.
Optional: Extra credits would be awarded if the hardware of the final prototype is built on a PCB.
- ☐ Collect the live data for at least 10 days.
- ☐ Perform data cleaning and analysis on the collected data.
- ☐ Calibration methodology should be clearly documented
- ☐ Error Checking and Diagnosis should be documented.

Project Grading

Initial proposal **15%**

Mid-project evaluation **35%**

You are expected to have the working prototype by this time.

Final evaluation **50%**

Working prototype + Data collection and analysis

Bonus (for optional PCB) **10%**